

Program: Diploma in Electrical Engineering & Electric Vehicle Technology	
Course Code: 2411	Course Title: Introduction to Electric Vehicles & Elementary Theory
Semester: 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:3 T:0 P:0)	Periods per semester: 45

Course Objectives:

- To know the history, development, classification, and basic terminology of automobile vehicles.
- To understand the working of Diesel and Petrol Engines, development of electric vehicles and appreciate the use EVs.
- To recognize lighting, heating and chemical effects of electric current.
- To provide knowledge of static and magnetic effects of electric current.

Course Prerequisites:

Topic	Course code	CourseName	Semester
Knowledge of basic science		Applied Physics I & II	1, 2

Course Outcomes:

On completion of the course, the student will be able to:

COs	Description	Duration (Hours)	CognitiveLevel
CO1	Summarize different classification of vehicles and terminology of Chassis specifications.	10	Understanding
CO2	Appreciate the use of electric vehicles.	11	Applying
CO3	Summarize the luminous, heating, and chemical effects of electric current.	11	Understanding
CO4	Classify the storage batteries and electrical engineering materials	11	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize different classification of vehicles and terminology of Chassis specification.		
M1.01	Explain the History of Automobile Vehicles in India and Abroad	3	Understanding
M1.02	Outline the various classifications of Vehicles	2	Understanding
M1.03	Outline different Vehicle layouts	3	Understanding
M1.04	Explain various terminology of vehicle specification	2	Understanding
Contents: History And Classification Automobile Vehicles - Definition - History and development in Abroad and in India before& after independence - Different Indian and foreign Automobile manufacturing concerns of two - wheeler, three-wheeler, LMV and HTV - their products and plants location. Classification of Automobile – Major systems of an Automobile. Chassis - definition, chassis layout of two, three & four – wheeler with major components – definition of : wheelbase - wheel track - overall length - front overhung - rear overhung height of CG point – ground clearance – gross weight and kerb weight.			
CO2	Appreciate the use of electric vehicles.		
M2.01	Understand the working principle of petrol & diesel engines	2	Understanding
M2.02	Understand the environmental impact of fossil fueled engines.	2	Understanding
M2.03	Explain the background & development of electric vehicles.	2	Applying

M2.04	List the various types of electric vehicles and their components.	3	Understanding
M2.05	Draw the layout of different electric vehicles.	2	Applying
	Series Test-I	1	
Contents: Development of EVs: Heatengines - Definition - types - engine terminology- classification of IC engineswith respect to different parameters, Two stroke & four stroke SI engines- working,Two stroke & Four stroke CI engines- working, comparison of SI and CI engines -comparison of Two stroke and Four stroke engines - emission characteristics of diesel and petrol engines- development of electric vehicles in India and abroad - types of EVs-components-layout-meritsofEVs-comparisonbetweenconventionalalandelectricvehicles.			
CO3	Summarize the luminous, heating, and chemical effects of electric current.		
M3.01	Explain the basic concepts of various lamps for residential and commercial purposes	3	Understanding
M3.02	Summarize the basic principle of electric heating and modes of heat transfer.	3	Understanding
M3.03	Classify various methods of electric heating.	3	Understanding
M3.04	Summarize the basic principle of electrolysis and familiarize the applications of electrolysis	2	Understanding
Contents: Electrical Lamps – types – basic diagram – working – incandescent –fluorescent–high pressure & low - pressure sodium vapour - LED -compact fluorescent – inductionlamp Electric Heating- Basic principle - advantages - requirements of good heating element - modes of heat transfer- methods of electric heating – principle -operation – resistance heating – direct – indirect – induction – direct core–dielectric (principle of operation only). Electrolysis – basic principle - Faraday’s laws- brief idea of applications of electrolysis.			
CO4	Classify the storage batteries and electrical engineering materials		
M4.01	Classify Storage batteries and summarize various terms related.	3	Understanding

M4.02	Identify the parts of lead acid batteries and its care and maintenance.	3	Understanding
M4.03	Illustrate the concept of series and parallel connection of battery.	3	Understanding
M4.04	Classify the electric and magnetic materials	2	
	SeriesTest-II	1	
Contents Storage battery- cell – battery - primary and secondary batteries - battery ratings –efficiency - ampere hour efficiency - watt hour efficiency - lead acid batteries-identificationof parts (brief idea only) - electrical Specifications – voltage – capacity - factors affectingcapacity - indications of a fully charged cell - applications - care and maintenance – lithium -ion battery - basic idea – applications - series and parallel connections of batteries - list themethods ofcharging. Electric materials– classification – properties - examples- conducting materials -insulating Magnetic materials- Ferromagnetic - Diamagnetic- Paramagnetic (properties & examples only)			

Text / Reference

T/R	Book Title/Author
T1	Ehsani, M. Modern Electric, Hybrid Electric and Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press
T2	Electric vehicle Technology -Prof.Sunil.R.Pawar-Notion press publication.
T3	Electric vehicle engineering-Per Enge,NickEnge,StephenZoepf-Mc Graw Hill.
T4	Automobile Engg.- Vol-II- DrKirpal Singh-Standard Publishers Distributors.
T5	Theraja, B. L., Electrical Technology Vol – I, S. Chand Publications, New Delhi, 2015, ISBN: 9788121924405
T6	Theraja, B. L. ;Theraja, A. K., A Text Book of Electrical Technology Vol-III, S. Chand and Co. Ramnagar, New Delhi, ISBN :9788121924900
R1	V.K.Mehta,Rohit Mehta Basic Electrical Engineering S. Chand Publications, New Delhi,
R2	Jegathesan, V., Basic Electrical and Electronics Engineering, Wiley India, New Delhi, 2015, ISBN : 97881236529513
R3	Chan C. C. and K. T. Chau, Modern Electric Vehicle Technology, Oxford Science Publication,
R4	Lechner G. and H. Naunheimer, Automotive Transmissions: Fundamentals, Selection,Design and Application, Springer
R5	Rashid, M. H. Power Electronics: Circuits, Devices and Applications, 3rd edition, Pearson,

R6	Moorthi, V. R. Power Electronics: Devices, Circuits and Industrial Applications, Oxford University Press
R7	Krishnan, R. Electric motor drives: modelling, analysis, and control, Prentice Hall 11. Krause, O. P. ; C. Wasynczuk, S. D. Sudhoff, Analysis of electric machinery, IEEE Press

Online Resources

Sl. No.	Website Link
1	https://nptel.ac.in/courses/108105112/
2	https://nptel.ac.in/courses
3	https://nptel.ac.in/courses/108/103/108103009/
4	https://onlinecourses.nptel.ac.in/noc20_ee99/preview
5	https://energyeducation.ca/encyclopedia/Main_Page
6	https://www.evacad.com
7	https://www.youtube.com/watch?v=UgtjRob5qMg
8	https://youtu.be/3E1SXXG7VkQk?si=grGF9k7DbWxkHatQ